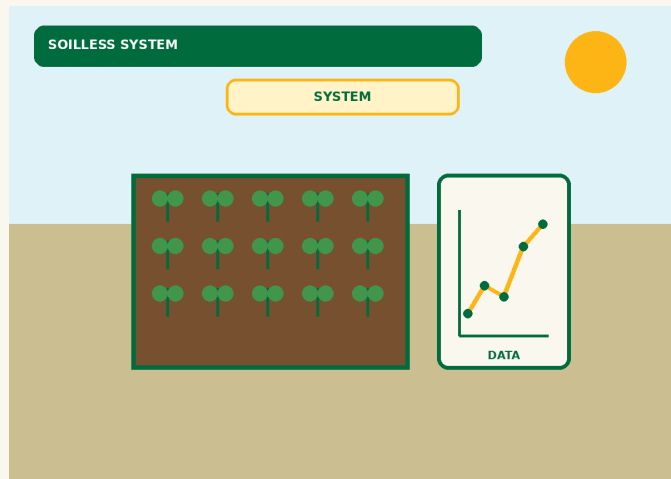




SYSTEM

SOILLESS SYSTEM

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 741 • SOILLESS SYSTEM

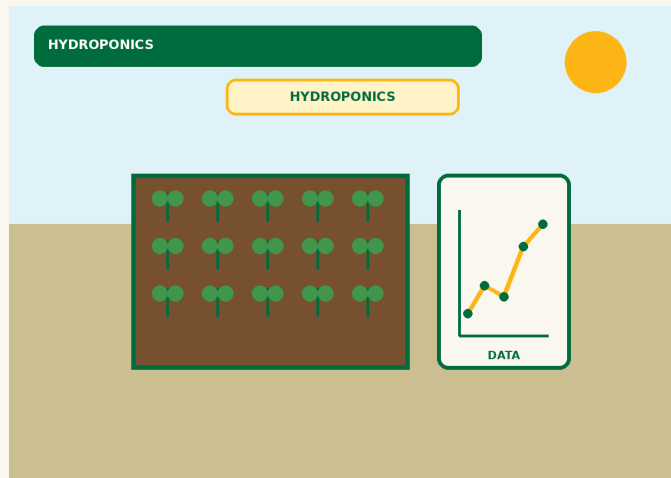
CONCEPT

A soilless system supplies roots with physical support, water, oxygen, and mineral nutrients without ordinary field soil.

INVESTIGATION TASK

Identify all major system components.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



HYDROPONICS

HYDROPONICS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 742 • HYDROPONICS

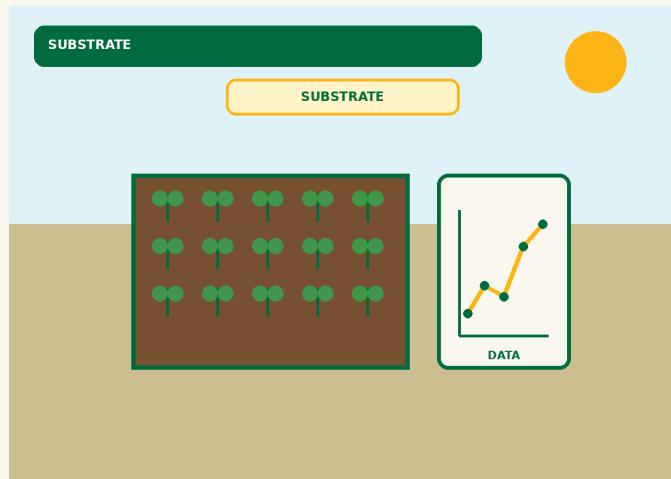
CONCEPT

Hydroponics is plant production in nutrient solution with or without an inert supporting medium.

INVESTIGATION TASK

Classify the school system by its root-support method.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SUBSTRATE

SUBSTRATE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 743 • SUBSTRATE

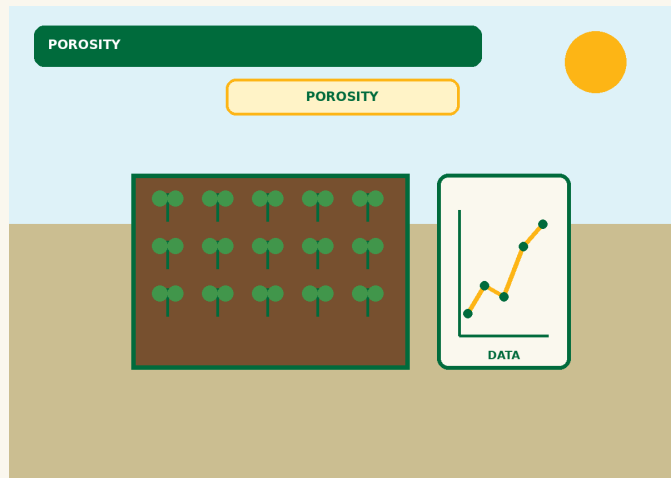
CONCEPT

A substrate is a soilless material used to support roots and influence water and air availability.

INVESTIGATION TASK

Compare two substrate properties.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



POROSITY

POROSITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 744 • POROSITY

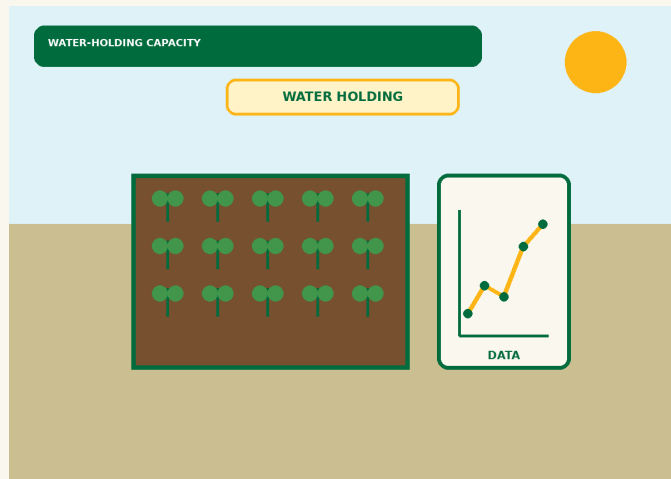
CONCEPT

Porosity is the fraction of substrate volume occupied by pores that can contain air or water.

INVESTIGATION TASK

Interpret a simplified pore-space diagram.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



WATER HOLDING

WATER-HOLDING CAPACITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 745 • WATER-HOLDING CAPACITY

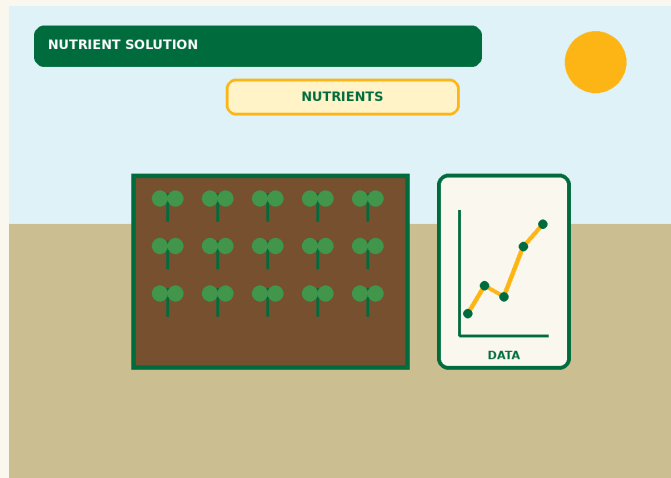
CONCEPT

Substrate water-holding capacity influences how much solution remains available between irrigation events.

INVESTIGATION TASK

Compare drainage from two media.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



NUTRIENTS

NUTRIENT SOLUTION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 746 • NUTRIENT SOLUTION

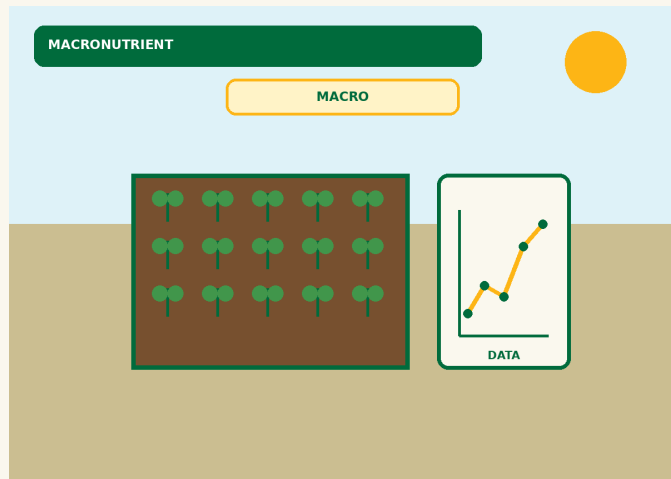
CONCEPT

Nutrient solution contains dissolved mineral ions supplied to support plant growth.

INVESTIGATION TASK

Identify the reservoir and delivery pathway.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MACRO

MACRONUTRIENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 747 • MACRONUTRIENT

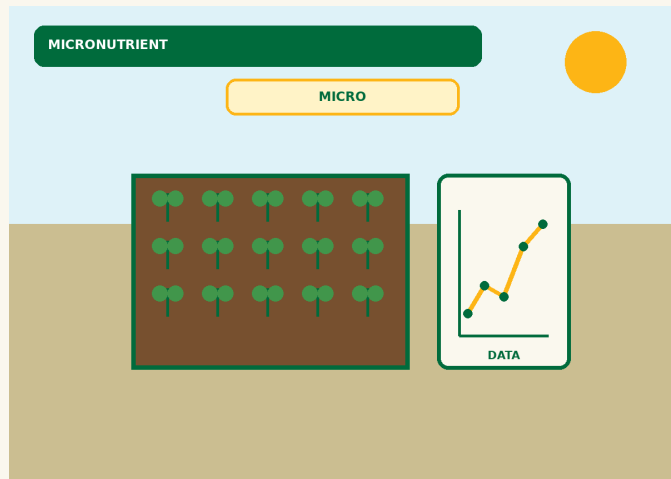
CONCEPT

Macronutrients are essential elements required by plants in relatively large quantities, including nitrogen, phosphorus, and potassium.

INVESTIGATION TASK

Match selected macronutrients with basic plant functions.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MICRO

MICRONUTRIENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 748 • MICRONUTRIENT

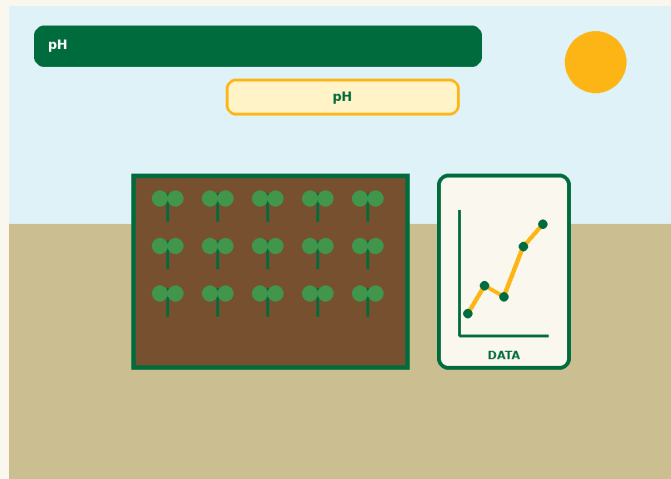
CONCEPT

Micronutrients are essential elements required in smaller quantities, such as iron, zinc, and boron.

INVESTIGATION TASK

Identify micronutrients from a provided list.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



pH

pH

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

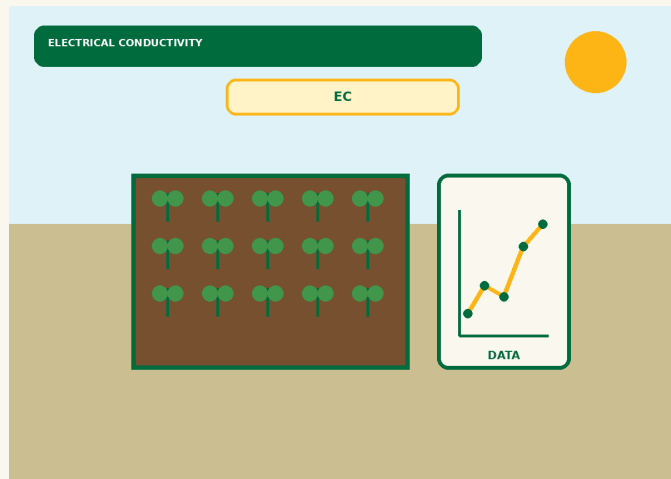
CONCEPT

Solution pH influences nutrient chemistry and availability to roots.

INVESTIGATION TASK

Compare a measured value with the target range.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EC

ELECTRICAL CONDUCTIVITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 750 • ELECTRICAL CONDUCTIVITY

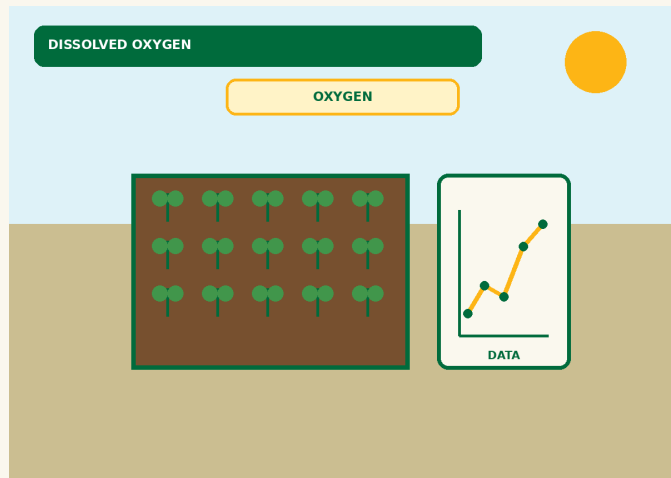
CONCEPT

Electrical conductivity indicates the solution's ability to conduct electricity and is related to dissolved ion concentration.

INVESTIGATION TASK

Interpret two EC readings.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



OXYGEN

DISSOLVED OXYGEN

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 751 • DISSOLVED OXYGEN

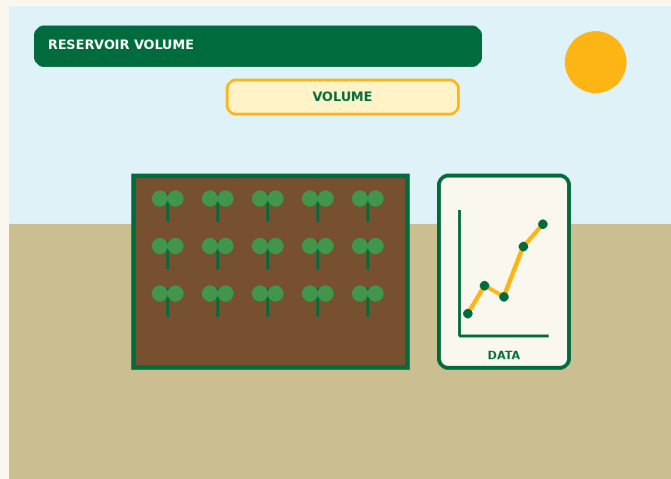
CONCEPT

Dissolved oxygen supports root respiration in water-based root zones.

INVESTIGATION TASK

Explain why stagnant solution can be problematic.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



VOLUME

RESERVOIR VOLUME

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 752 • RESERVOIR VOLUME

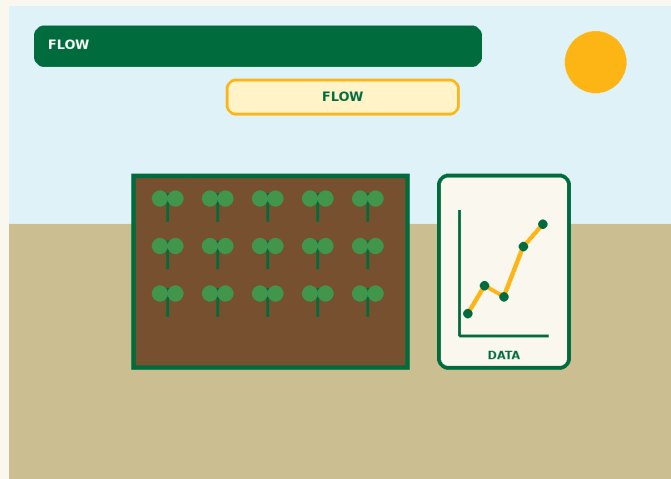
CONCEPT

Reservoir volume affects nutrient-solution capacity, concentration stability, and management frequency.

INVESTIGATION TASK

Calculate volume from tank dimensions or markings.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FLOW

FLOW

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

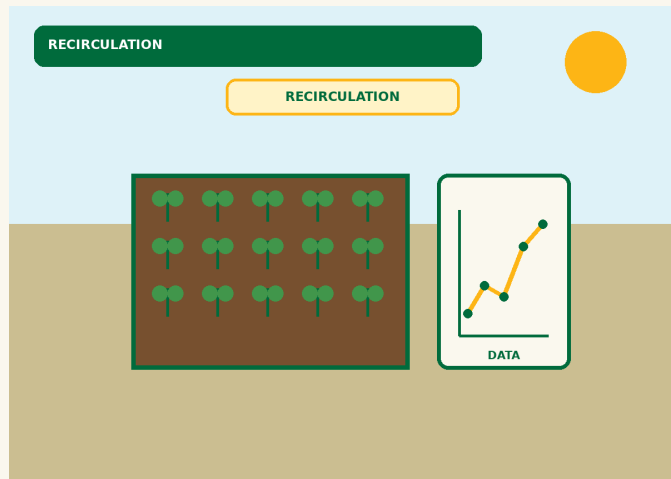
CONCEPT

Solution flow transports water and dissolved nutrients through the system.

INVESTIGATION TASK

Measure or compare flow at two points.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RECIRCULATION

RECIRCULATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 754 • RECIRCULATION

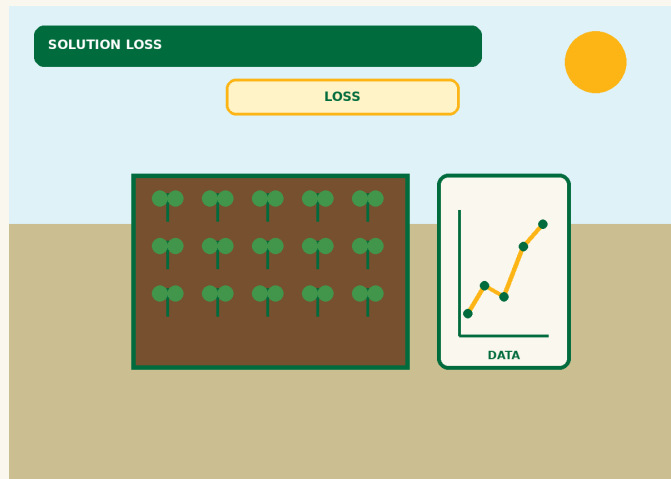
CONCEPT

Recirculation returns drainage or flowing solution to the reservoir for reuse.

INVESTIGATION TASK

Trace one complete circulation loop.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LOSS

SOLUTION LOSS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 755 • SOLUTION LOSS

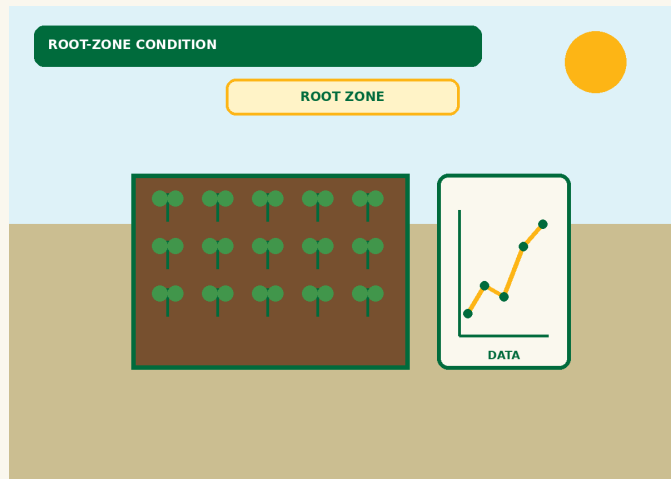
CONCEPT

Solution can leave the system through plant uptake, evaporation, leaks, drainage, or maintenance.

INVESTIGATION TASK

Identify likely loss pathways.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ROOT ZONE

ROOT-ZONE CONDITION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 756 • ROOT-ZONE CONDITION

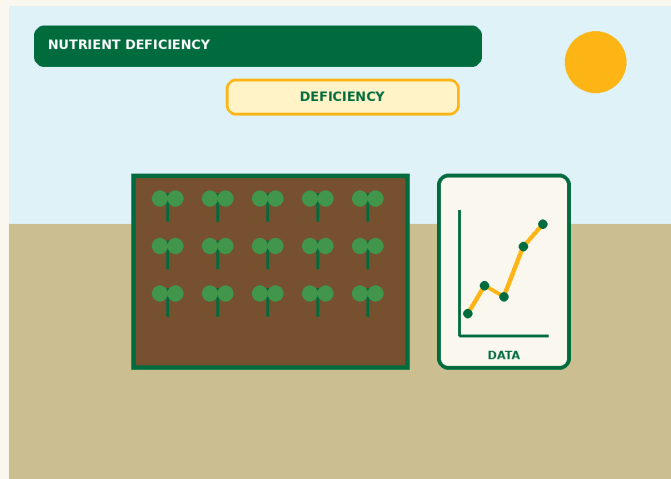
CONCEPT

Root-zone condition includes moisture, oxygen, temperature, solution chemistry, and root health.

INVESTIGATION TASK

Complete a root-zone observation checklist.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DEFICIENCY

NUTRIENT DEFICIENCY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 757 • NUTRIENT DEFICIENCY

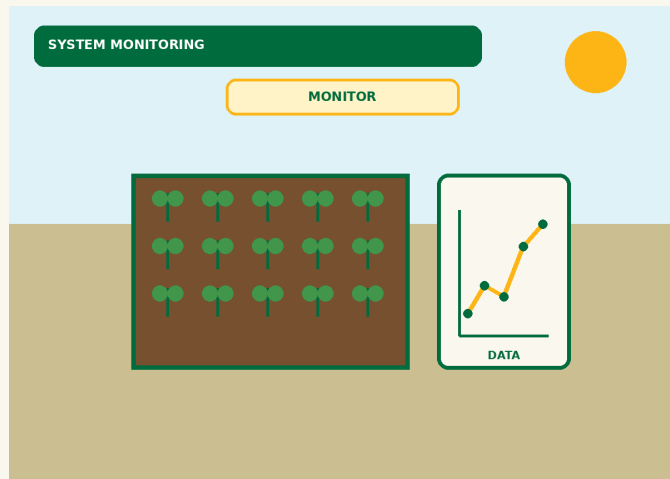
CONCEPT

Nutrient deficiency symptoms can indicate inadequate nutrient supply but must be interpreted cautiously because other stresses may look similar.

INVESTIGATION TASK

Compare symptoms using teacher-provided references.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MONITOR

SYSTEM MONITORING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 758 • SYSTEM MONITORING

CONCEPT

Monitoring combines pH, EC, solution level, flow, root condition, and crop measurements.

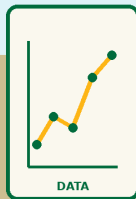
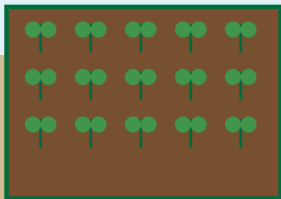
INVESTIGATION TASK

Complete a monitoring record.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE

TROUBLESHOOTING

TROUBLESHOOT



TROUBLESHOOT

TROUBLESHOOTING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 759 • TROUBLESHOOTING

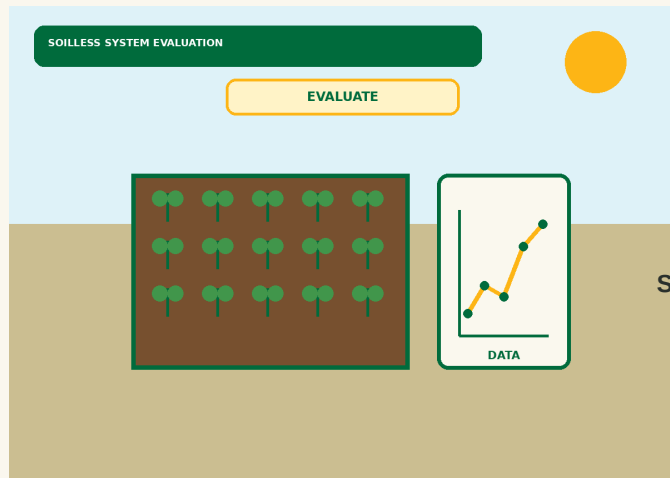
CONCEPT

Troubleshooting uses symptoms, measurements, and system checks to identify likely causes before corrective action.

INVESTIGATION TASK

Diagnose a simple teacher-provided scenario.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EVALUATE

SOILLESS SYSTEM EVALUATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 760 • SOILLESS SYSTEM EVALUATION

CONCEPT

Evaluation connects solution management, root-zone condition, crop response, resource use, and operational reliability.

INVESTIGATION TASK

Recommend one monitored improvement.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE